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PROFILE

MSc Data Science and BSc Mathematics graduate with a strong foundation in machine learning, statistics, data analysis, and predictive modelling. Experienced in analysing complex datasets, building machine learning solutions, developing dashboards, and delivering actionable insights through data-driven decision-making. Proficient in Python, SQL, Excel, Power BI, and modern machine learning techniques, with academic experience spanning NLP, computer vision, forecasting, and AI ethics. Strong analytical thinker with excellent problem-solving skills, attention to detail, and a keen interest in responsible AI, emerging technologies, and business-focused analytics.

EDUCATION

MSc. Data Science with Advanced Research (Distinction), University of Hertfordshire, Hatfield, UK (Jan 2023 – Mar 2025)

- Developed advanced skills in machine learning, statistical modelling, Python programming, SQL, and large-scale data analytics.
- Thesis: Predicting Movie Ratings and Analysing Feature Importance Using Tags and Genres
- Relevant Modules: Statistical Modelling, Python, SQL, Data Analysis, Data Visualisation, Time Series Analysis, Machine Learning, NLP, LLM, AI Ethics.

BSc. Mathematics (Distinction), St. Berchmans College, Kerala, India (Jun 2019 – Sept 2022)

- Strong foundation in Probability, Statistics, Economics, Hypothesis Testing, Calculus, Linear Algebra, Differential Equations, and Real Analysis.
- Attained rigorous quantitative reasoning and problem-solving skills applicable across economic, financial, and statistical data analysis.

TECHNICAL SKILLS

- **Data Analytics & ML:** Data Querying, Data Cleaning, Data Validation, Exploratory Data Analysis, Statistical Analysis, Feature Engineering, Time Series Analysis, Predictive Modelling, Model testing & validation, NLP, LLMs.
- **Programming:** Python (Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch), SQL, R, Git.
- **Reporting & Visualisation:** Excel (Pivot Tables, Lookups, VBA), Dashboard Development, Power BI, Tableau, Streamlit, KPI Reporting.
- **Cloud & Other Tools:** Microsoft Azure, Azure AI, AWS, Production Support, GitHub, Microsoft Office Suite, LaTeX.

PROJECTS

Predicting Movie Ratings and Analysing Feature Importance Using Tags and Genres

- Implemented predictive models in Python to analyse relationships between user-generated tags, genres, and movie ratings within a large-scale dataset.
- Performed exploratory data analysis and feature engineering on large datasets.
- Applied statistical analysis to identify rating patterns and user behaviour trends
- Reviewed model performance using statistical error metrics and visual analysis.

Financial Market Forecasting

- Prepared forecasting models using historical market data to analyse pricing trends and market behaviour.
- Conducted data validation, cleaning, anomaly detection, time-series analysis and data preprocessing to identify market trends and predictive factors.
- Examined model performance using statistical error metrics against baseline forecasting methods and sensitivity analysis to evaluate forecasting reliability.
- Built Power BI dashboards to communicate financial trends, model outputs, and decision-support insights.

Fine-Tuning DistilBERT for Toxicity Classification

- Fine-tuned a DistilBERT transformer model using TensorFlow(Python) to detect toxic and harmful online content.
- Prepared tokenisation, text preprocessing, and model optimisation techniques for NLP classification.
- Evaluated model performance using ROC-AUC score and other classification metrics.
- The method showed a high recall for toxic comments while occasionally labelling non-toxic comments as toxic,

prioritising identifying all potential toxicity over preventing false positives.

Image Segmentation with Deep Learning

- Tested a convolutional neural network for image segmentation using a curated subset of the COCO 2017 dataset.
- Devised image preprocessing, augmentation, and training techniques to improve model performance.
- Critiqued segmentation quality using Intersection over Union (IoU) metrics.
- Visualised predicted segmentations to assess object boundary detection in complex scenes.

PROFESSIONAL EXPERIENCE

Night Auditor and Guest Service Assistant | Holiday Inn Express (IHG Brand), Watford, UK **(June 2023 – Present)**

- Reconcile daily financial and operational records, ensuring accuracy across revenue, occupancy, and transaction data.
- Audit discrepancies across systems, identifying root causes and resolving data inconsistencies.
- Produce financial and operational reports, supporting performance monitoring, trend analysis and management decision-making.
- Investigate monthly, quarterly and annual occupancy, revenue and customer review using dashboards to identify patterns, anomalies and business performance insights.
- Manage confidential customer and financial information with accuracy and professionalism.
- Collaborate with finance, management, housekeeping, and front-office teams to resolve issues and improve reporting accuracy and workflow efficiency.
- Automated welcome letter generation for IHG members using Excel and Word, incorporating tier-level benefits and alphabetical ordering, replacing a manual system.
- Operate within a highly regulated, accuracy-focused environment requiring strong attention to detail, analytical thinking and adherence to reporting deadlines.

Mathematics Tutor | Universal Tuition Centre, Kerala, India **(May 2021 - Jan 2023)**

- Deliver structured mathematics and statistics tuition, developing strong analytical and quantitative reasoning skills.
- Communicate complex concepts in a clear and structured manner, strengthening stakeholder communication skills.
- Handle multiple student schedules and learning objectives, demonstrating organisation and discipline.

PROFESSIONAL CERTIFICATIONS

- Microsoft Certified: Azure AI Engineer Associate
- Microsoft Certified: Azure AI Fundamentals

VOLUNTEERING & ACTIVITIES

- Active committee member of the Physics, Astronomy and Mathematics Society, Sim Racing Society, and Anime Society at the University of Hertfordshire.
- Monitored events and collaborated with diverse teams to deliver successful student activities.
- Led the Rubik's Cube competition during college events, demonstrating leadership, organisation, and teamwork skills.
- Achieved first place in several mathematics competitions during school and university, demonstrating strong mathematical skills and problem-solving abilities.

ADDITIONAL INFORMATION

Languages - Fluent in *English*, Hindi, Malayalam.

Hobbies and Interests - AI | Sports Analysis | Rubik's Cubes | Playing Keyboard | Football | Cricket | Motorsports | Sim Racing | Movies | Music